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First Name(s):														

Department of Mathematics and Applied Mathematics
 MAM2000W: 2LA (Linear Algebra)

Test 1 Time : 1.5 hours	10 March 2020 Maximum marks: 51 Full marks: 50
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- This question paper consists of 8 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π , e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
10	6	11	7	8	4	5	51

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Question 1. [10 points] Let $A = \begin{pmatrix} 0 & 8 & 2 \\ 4 & 14 & 0 \\ -8 & -12 & 4 \end{pmatrix}$.

(a) Find elementary matrices E_1 , E_2 and E_3 and a matrix U in row echelon form such that $E_3E_2E_1A = U$.

Do Gaussian elimination on A to obtain its row echelon form:

$$\begin{pmatrix} 0 & 8 & 2 \\ 4 & 14 & 0 \\ -8 & -12 & 4 \end{pmatrix} \sim \begin{pmatrix} 4 & 14 & 0 \\ 0 & 8 & 2 \\ -8 & -12 & 4 \end{pmatrix} \begin{array}{l} R_2 \\ R_1 \end{array}$$

$$\sim \begin{pmatrix} 4 & 14 & 0 \\ 0 & 8 & 2 \\ 0 & 16 & 4 \end{pmatrix} \quad R_3 + 2R_1$$

$$\sim \begin{pmatrix} 4 & 14 & 0 \\ 0 & 8 & 2 \\ 0 & 0 & 0 \end{pmatrix} \quad R_3 - 2R_2$$

Therefore we can take

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$$

and $U = \begin{pmatrix} 4 & 14 & 0 \\ 0 & 8 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

(There are also other correct solutions.)

(b) Find the the set of solutions of the equation $A\mathbf{x} = \mathbf{0}$.

x_3 is a free variable (since there is no pivot in the third column). Put $x_3 = t$, where $t \in \mathbb{R}$. Then the second row yields $x_2 = -\frac{1}{4}x_3 = -\frac{1}{4}t$ and the first row $2x_1 = -7x_2$, or $x_1 = -\frac{7}{2}x_2 = \frac{7}{8}t$. Hence the solutions are all of the form

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = t \begin{pmatrix} 7 \\ -2 \\ 8 \end{pmatrix}, t \in \mathbb{R}.$$

(c) Answer the following questions, giving reasons for your answers:

(i) Can A be written as a product of elementary matrices?

No. The system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions, from (b). Hence A is singular, by definition, and it can therefore not be written as a product of elementary matrices.

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(ii) Is the matrix U non-singular?

The system $U\mathbf{x} = \mathbf{0}$ has the free variable x_3 , and therefore has infinitely many solutions. Hence U is singular.

Question 2. [6 points] In this question, you are given the augmented matrices of two linear systems, already in row echelon form. Write down the **solution set** of the system in each case.

(a)

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 9 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

The solution set is $\left\{ \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} \right\}$.

(b)

$$\left(\begin{array}{cccc|c} 1 & 3 & 5 & 7 & 2 \\ 0 & 0 & 1 & 1 & 4 \end{array} \right)$$

The solution set is $\left\{ \begin{pmatrix} -18 \\ 0 \\ 4 \\ 0 \end{pmatrix} + s \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\}$

Question 3. [11 points] Let

$$A_1 = \begin{pmatrix} 2 & 0 & -4 \\ 0 & -3 & 0 \\ 0 & 0 & -7 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix},$$

$$A_4 = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 7 \end{pmatrix}, \quad A_5 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_6 = \begin{pmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

In each of (a) to (g) below, a description of a matrix is given. In each case, choose **one** matrix from the list above satisfying the description given. Write **only one of** $A_1, A_2, A_3, A_4, A_5, A_6$ in the appropriate block on the answer sheet. If it is impossible to find a matrix satisfying the description given, write “Impossible”. You may use a matrix more than once, or not at all.

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- (a) A matrix that is not row-equivalent to I_3 .

A_4

- (b) A symmetric matrix in row echelon form.

A_5

- (c) A singular elementary matrix.

Impossible

- (d) An upper triangular matrix which is invertible. A_1 or A_5

- (e) An elementary matrix that is not in row echelon form. A_2

- (f) A matrix which can be written as the product of two elementary matrices, neither of which is an identity matrix.

A_2

Also give the two elementary matrices, in the correct order.

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

OR

A_3

Also give the two elementary matrices, in the correct order.

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}$$

- (g) A matrix which is not elementary and not in row echelon form, but can be written as the product of an elementary matrix and a matrix in row echelon form.

A_4

Give the two matrices in the product, in the correct order.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{7}{2} & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & -3 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

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OR

A_6

Also give the two elementary matrices, in the correct order.

$$\begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Question 4. [7 points] Consider the statement:

For every $n \times n$ matrix A , if $A^2 = A$ and $A \neq I_n$, then A is not invertible. (*)

(a) Write down the converse of the statement (*).

The converse is: For every $n \times n$ matrix A , if A is not invertible, then $A^2 = A$ and $A \neq I_n$.

(b) Write down the contrapositive of the statement (*).

The contrapositive is: For every $n \times n$ matrix A , if A is invertible, then $A^2 \neq A$ or $A = I_n$.
(If you had “and” here in stead of “or”, your answer was incorrect; this should be clear since if $A = I_n$, then $A^2 = A$!)

(c) Prove or disprove the statement (*).

The statement (*) is true. The contrapositive of this statement is logically equivalent to the statement. We prove the contrapositive:

If A is invertible, then $A^2 \neq A$, or $A^2 = I_n$.

Let A be invertible and assume that

$$A^2 = A.$$

Multiply both sides of the equation (from the left) with A^{-1} (which exists since A is invertible). We obtain

$$A^{-1}A^2 = A^{-1}A.$$

Hence

$$A^{-1}(AA) = A^{-1}A.$$

By associativity of matrix multiplication, it follows that

$$(A^{-1}A)A = A^{-1}A.$$

This simplifies to

$$I_n A = I_n,$$

and hence

$$A = I_n.$$

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(d) Prove or disprove the converse of the statement (*).

The converse is false. A possible counter example is: $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ is not invertible, but $A \neq I_2$ and $A^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \neq A$.

Question 5. [8 points] Read the following result and its proof and then answer the questions following it.

Theorem 1. *Every elementary matrix is invertible, and its inverse is also an elementary matrix.*

Proof. Let E be an $n \times n$ elementary matrix.

Then $E = e(I_n)$, for some elementary row operation e . (1)

Let e^{-1} be the inverse operation of e . (2)

Then $E \cdot e^{-1}(I_n) = e(I_n) \cdot e^{-1}(I_n) = e(e^{-1}(I_n)) = I_n$ (3)

and $e^{-1}(I_n) \cdot E = e^{-1}(E) = e^{-1}(e(I_n)) = I_n$. (4)

It follows that E is invertible, (5)

and that E^{-1} is also an elementary matrix. (6)

□

(a) If e is an elementary row operation and A an $n \times n$ matrix, what does the symbol $e(A)$ stand for?

$e(A)$ is the matrix obtained by performing the elementary row operation e on the matrix A . (e is an elementary row operation; $e(A)$ is **matrix**, not an operation.)

(b) There is a matrix C such that $e(A) = CA$. How is the matrix C obtained?

$C = e(I_n)$.

(c) Use your answer to (b) to explain why we can say in (3) that

$$e(I_n) \cdot e^{-1}(I_n) = e(e^{-1}(I_n)).$$

If we take $A = e^{-1}(I_n)$, then by (b),

$$e(e^{-1}(I_n)) = e(A) = e(I_n)A = e(I_n)e^{-1}(I_n).$$

(d) Why can we say at (5) that E is invertible?

It follows from lines (3) and (4) that there is a matrix $B = e^{-1}(I_n)$ such that $EB = I_n = BE$. This shows that E is invertible.

(e) How does it follow at (6) that E^{-1} is elementary?

It follows from (d) that $E^{-1} = B = e^{-1}(I_n)$. But since e^{-1} is an elementary row operation, $e^{-1}(I_n)$ is an elementary matrix, by definition.

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(f) Can an elementary matrix be its own inverse? Give a reason for your answer.

Yes. An obvious example is the identity matrix. Alternatively, if e is the operation of interchanging two rows, then $e^{-1} = e$ and hence $(e(I_n))^{-1} = e^{-1}(I_n) = e(I_n)$.

(g) Prove or disprove: The inverse of the product of two elementary matrices is an elementary matrix.

The statement is false. As an example, $E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $F = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ are both elementary matrices, but $(EF)^{-1} = F^{-1}E^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & 0 \end{pmatrix}$, and this is not an elementary matrix. *The product of two elementary matrices are in general not elementary, but it could be. If you argued (correctly) that the inverse of the product of two elementary matrices is the product of two elementary matrices, it does therefore not follow that the latter product is necessarily not elementary. To make the point, it is easiest to give an example.*

Question 6. [4 points] For each of the following statements, say whether it is true or false. If it is true, prove it; if it is false, give a counter example and explain why it is one.

(a) (2 marks) The sum of two elementary matrices is always an elementary matrix.

False. As a counterexample,

$$E_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \text{ and } E_2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

are elementary matrices (both can be obtained from I_2 by using one elementary operation), but $E_1 + E_2 = \begin{pmatrix} 2 & 0 \\ 3 & 2 \end{pmatrix}$ is not an elementary matrix, since it cannot be obtained by performing just one elementary operation on I_2 .

(b) (2 marks) For any invertible matrices A and B , A^7B^5 is invertible.

True. The product of two invertible matrices is again invertible. Since A is invertible, $A^2 = AA$, $A^3 = AA^2, \dots, A^7 = AA^6$ are all invertible. In the same way it follows that B^5 is invertible. Then the product A^7B^5 is invertible.

Question 7. [5 points] For any $n \times n$ matrix $C = (c_{ij})$, the *trace* of C , denoted by $\text{tr}(C)$, is defined by

$$\text{tr}(C) = \sum_{i=1}^n c_{ii}.$$

(a) (2 marks) Prove that for any $n \times n$ matrices A and B ,

$$\text{tr}(AB) = \text{tr}(BA).$$

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Let $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$ and $D = (d_{ij})$, where $C = AB$ and $D = BA$. Then $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$ and $d_{ij} = \sum_{k=1}^n b_{ik}a_{kj}$, for $1 \leq i, j \leq n$. Hence

$$\begin{aligned}
 \text{tr}(AB) &= \sum_{i=1}^n c_{ii} \\
 &= \sum_{i=1}^n \sum_{k=1}^n a_{ik}b_{ki} \\
 &= \sum_{i=1}^n \sum_{k=1}^n b_{ki}a_{ik} \\
 &= \sum_{k=1}^n \sum_{i=1}^n b_{ki}a_{ik} \\
 &= \sum_{k=1}^n d_{kk} \\
 &= \text{tr}(BA)
 \end{aligned}$$

(b) (2 marks) Prove or disprove: There are $n \times n$ matrices A and B such that $AB - BA = I_n$.

The statement is false. We prove this by contradiction. Suppose there are two $n \times n$ matrices A and B such that $AB - BA = I_n$. Then it follows from (a) that

$$0 = \text{tr}(AB) - \text{tr}(BA) = \text{tr}(AB - BA) = \text{tr}(I_n) = n,$$

a contradiction. We have used the fact that if $C = (c_{ij})$ and $D = (d_{ij})$ are $n \times n$ matrices, then

$$\text{tr}(C - D) = \sum_{i=1}^n (c_{ii} - d_{ii}) = \sum_{i=1}^n c_{ii} - \sum_{i=1}^n d_{ii} = \text{tr}(C) - \text{tr}(D).$$

(c) (1 mark) Let A be any skew-symmetric matrix. Find $\text{tr}(A)$.

For an $n \times n$ skew-symmetric matrix $A = (a_{ij})$, we have $A = -A^T$, and so $a_{ij} = -a_{ji}$ for all i, j . In particular, $a_{ii} = -a_{ii}$, from which it follows that $a_{ii} = 0$ for $i = 1, 2, \dots, n$. Therefore $\text{tr}(A) = 0$.